Molecular Mechanism of the DUF1156 Anti-Phage Defense Protein Family (NF042963)

Summary

The DUF1156-containing anti-phage defense family **NF042963** is, at its enzymatic core, a **S-adenosyl-L-methionine (SAM)-dependent DNA methyltransferase**. Structure-based fold assignment of the family representative **A0A3B7MFS0** (*Thermosynechococcus sichuanensis* E542, 1002 aa) resolves a single, intact catalytic module embedded in the middle of the ~1000-residue chain and assigns it, with the highest possible confidence, to a **β-class, site-specific DNA cytosine-N4 methyltransferase (m4C; EC 2.1.1.113)**. Foldseek searches of the AlphaFold model against both AFDB-SwissProt and PDB100 return N4-cytosine-specific DNA methyltransferases (M.MjaII, M.MvaI, and the cyanobacterial M.AvaI) as the top hits at probability 1.0. The catalytic machinery is fully preserved: the Class I methyltransferase motif I glycine loop (FxGxG, F233–G237) and the diagnostic amino-methyltransferase catalytic motif IV (**DPPY**, D617–Y620) converge into a single Rossmann-fold SAM pocket in the model.

Critically, the protein carries **no nuclease fold of any kind** — there is no PD-(D/E)xK, HNH/His-Me finger, GIY-YIG, or m5C-type Pro-Cys motif anywhere in the sequence or structure. The only assignable catalytic fold across the entire 1002-aa chain is the central m4C methyltransferase. The defining **DUF1156 domain (PF06634)** itself is a small N-terminal module with no recognizable catalytic fold, and the C-terminal extension is likewise structurally novel and non-catalytic. Both flanking regions are therefore **accessory** — most plausibly serving DNA/target recognition and scaffolding roles, not catalysis.

The mechanistic conclusion is that this family defends against phages by **epigenetic self versus non-self discrimination**, in the mould of the **BREX / DISARM / restriction–modification** class of modification-dependent defense systems. The methyltransferase marks host DNA at a specific sequence motif via N4-cytosine methylation; unmethylated invading phage DNA lacks this self-mark and is excluded or restricted (with any nuclease/effector step supplied in *trans* by other genes of the defense locus, as in BREX). The single decisive confirmatory experiment is to build a DPPY catalytic-dead mutant (e.g., **D617A / Y620A**) and

show that it abolishes both N4-cytosine DNA methylation activity and phage protection. **Overall confidence: high** for the enzymatic identity (m4C DNA methyltransferase); **moderate-to-high** for the specific BREX/DISARM-style modification-dependent exclusion mechanism.

Key Findings

Finding 1 — The protein is a SAM-dependent DNA amino-methyltransferase with an intact catalytic DPPY active site

The core enzymatic assignment rests on both sequence-motif mapping and 3D geometry. InterPro maps an **S-adenosyl-L-methionine-dependent methyltransferase superfamily domain (IPR029063)** to residues 157–287 of A0A3B7MFS0, within the broader NF042963 family signature (IPR049953, residues ~26–996). Within that window sits the canonical **Class I methyltransferase motif I** — the glycine-rich SAM-binding loop with the signature FxGxG pattern, realized here as **F233-C-G-S-G237** (context `...ADTFCGSGQIPFEA...`). Downstream, the diagnostic **amino-methyltransferase catalytic motif IV, DPPY**, is intact at **D617-P618-P619-Y620** (context `ITDPPYGD`). The DPPY tetrad is the hallmark of adenine-N6 and cytosine-N4 (amino) methyltransferases; its aspartate orients the exocyclic amino group of the target base and its aromatic tyrosine stacks against the flipped-out base, so its presence in undegraded form is strong evidence of an active amino-MTase.

Importantly, a systematic search for competing catalytic modules came up empty: there is **no PD-(D/E)xK, no HNH, no GIY-YIG, and no m5C-type Pro-Cys (PCQ) catalytic motif** anywhere in the chain. This means the module is *exclusively* a methyltransferase and specifically an **amino-methyltransferase** (targeting an exocyclic amino group), not a 5-methylcytosine enzyme and not a nuclease.

The AlphaFold model (**AF-A0A3B7MFS0-F1**, mean pLDDT 86.8, with 89% of residues above pLDDT 70) confirms the functional geometry. The motif I glycine loop and the catalytic DPPY, though ~380 residues apart in sequence, **converge into one pocket** in 3D: the Cα–Cα distance between G235 and D617 is only **8.1 Å**, and the entire motif I stretch (residues 230–240) lies within 12 Å of catalytic Asp617. This is exactly the spatial arrangement expected of a Rossmann-fold SAM-methyltransferase active site, where the SAM cofactor (bound by motif I) sits adjacent to the target-base catalytic residues (motif IV). Confidence over the catalytic region is high (motif I region pLDDT ≈ 94.3; DPPY pLDDT ≈ 82.5), so the geometry is trustworthy rather than a low-confidence artifact. **The catalytic residues are present, not degraded.**

Finally, this analysis relocated the DUF1156 domain itself: **IPR009537 / PF06634 maps to a small, separate N-terminal domain (residues ~30–90)**, well outside the catalytic module. The name "DUF1156" therefore labels an accessory sub-region, not the catalytic center of the protein.

Finding 2 — Foldseek assigns the catalytic module to a β-class N4-cytosine DNA methyltransferase (m4C, EC 2.1.1.113)

Sequence motifs establish "amino-MTase"; structure decides the target base. A **Foldseek** search of the AlphaFold model against AFDB-SwissProt and PDB100 returned, as its highest-confidence hits (**probability = 1.0**), experimentally annotated **N4-cytosine-specific DNA methyltransferases**:

Foldseek target	Protein	E-value	Query span	Annotated activity
Q58843	Type II MTase **M.MjaII**	4.7×10^{-7}	212–744	m4C DNA MTase (EC 2.1.1.113)
P14244	Type II MTase **M.MvaI**	7.7×10^{-5}	212–742	m4C DNA MTase (EC 2.1.1.113)
P0A461 / P0A462	Type II MTase **M.AvaI (cyanobacterial)**	1.6×10^{-3}	207–742	m4C DNA MTase (EC 2.1.1.113)

Each of these enzymes catalyzes the reaction *a 2′-deoxycytidine in DNA + S-adenosyl-L-methionine → an N(4)-methyl-2′-deoxycytidine in DNA + S-adenosyl-L-homocysteine* (**EC 2.1.1.113**). The structural alignment spans query residues ~207–744, covering **both** motif I (FCGSG at 233) and catalytic motif IV (DPPY at 617) in one contiguous fold. That single, continuous span containing both motifs — with a large inserted target-recognition region between them — is the defining topology of the **β-class amino-methyltransferases**, in which the motif order along the chain places the catalytic motif IV before the SAM-binding-region-derived recognition insertion, distinguishing β-class enzymes from α- and γ-class amino-MTases.

Two additional details sharpen the call. First, the PDB100 hits (probability ≈ 0.99), which align only the ~586–745 catalytic subdomain, are generic **Class I SAM/O-methyltransferase Rossmann folds** — consistent with, and nested within, the amino-MTase assignment. Second, **M.AvaI is itself a cyanobacterial enzyme** (from *Nostoc/Anabaena*), the same phylum as the representative *Thermosynechococcus* — a phylogenetically coherent top match. Notably, **no m6A-specific (adenine) or m5C (Pro-Cys) enzyme ranked above the m4C hits**, so the structure specifically favors cytosine-N4 over the alternative amino-MTase target (adenine-N6).

Finding 3 — DUF1156 and the C-terminal extension have no assignable catalytic fold (accessory modules)

To test whether any part of the protein *other* than the central module could be catalytic, per-region Foldseek searches were run against AFDB-SwissProt and PDB100. The **N-terminal region (residues 1–210)** — which contains the DUF1156 domain at residues 30–90 — returned **zero significant structural hits**. The **C-terminal region (residues 745–1002)** likewise returned no real hit (best probability 0.06, E-value 8.3 — spurious). By contrast, the **central region (207–744) matched N4-cytosine DNA methyltransferases at probability 1.0**.

The interpretation is unambiguous: the **only assignable catalytic fold in the entire 1002-aa chain is the central m4C methyltransferase**. The DUF1156 domain and the C-terminal extension are structurally novel and non-catalytic. In the context of a site-specific DNA methyltransferase, non-catalytic flanking domains typically function as **target-recognition domains (TRDs), DNA-binding/scaffolding modules, or partner-protein interaction surfaces**. The DUF1156 module is therefore best described as **accessory** — it contributes recognition/specificity or complex assembly, not the chemistry of methyl transfer.

Finding 4 — Defense mechanism: methylation-based self/non-self discrimination, not nucleic-acid degradation

Combining Findings 1–3, the protein's *sole* enzymatic activity is DNA methylation, and it lacks any nuclease. A large (~1000 aa) multidomain protein whose only catalytic output is site-specific DNA methylation is architecturally the signature of the **epigenetic, modification-based anti-phage systems** — **BREX** (BrxX/PglX m6A methyltransferase) and **DISARM** (DrmMII methylase) — rather than of a classical restriction–modification (R-M) system in which a paired endonuclease performs the killing step. In BREX and DISARM, site-specific methylation of host DNA serves as a **self-marker**: incoming phage DNA, lacking the host methylation pattern, is recognized as non-self and excluded or restricted. This is mechanistically distinct from R-M cleavage and consistent with the absence of any nuclease domain in NF042963: **any effector/nuclease step would be supplied in *trans*** by other genes of the defense operon.

Mechanistic Model / Interpretation

Domain architecture of A0A3B7MFS0 (1002 aa)

```
N ──[ DUF1156 ]──────[ β-class m4C DNA methyltransferase ]──────────[ C-term ext ]── C
1    30    90  157         233            617                  744    745          1002
     └─PF06634─┘  │      FxGxG(I)        DPPY(IV)               │      └─ no fold ─┘
     accessory    └──── IPR029063 SAM-MTase / Rossmann fold ──┘        accessory
     (recognition   Foldseek → M.MjaII / M.MvaI / M.AvaI (EC 2.1.1.113)  (scaffold /
      / scaffold)   probability = 1.0                                    partner IF)

Active-site geometry (AlphaFold model AF-A0A3B7MFS0-F1, mean pLDDT 86.8):
  G235(Cα) ── 8.1 Å ── D617(Cα)      motif I loop within 12 Å of catalytic Asp617
  → single Rossmann SAM pocket, canonical amino-MTase active site
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Enzyme identity and reaction

- **Enzyme class:** Site-specific DNA **cytosine-N4 methyltransferase** (m4C), β-class amino-MTase.
- **EC number: 2.1.1.113**
- **Reaction:** 2′-deoxycytidine (in a specific DNA recognition sequence) + SAM → N4-methyl-2′-deoxycytidine + S-adenosyl-L-homocysteine.
- **Catalytic residues (intact):** motif I FxGxG (F233–G237, SAM binding); motif IV **DPPY (D617–Y620)**, orienting the exocyclic N4 amino group of the flipped target cytosine.

How the activity confers phage defense

```
       HOST GENOME                            INVADING PHAGE DNA
───(m4C)C───site───C(m4C)───          ───C───site───C───   (unmethylated)
         │  "self" mark                         │  no self mark
         ▼                                      ▼
tolerated / replicated normally         recognized as NON-SELF
                                          → excluded / restricted
                                          (effector supplied in trans
                                           by other operon genes, BREX-like)
```

The methyltransferase writes an N4-cytosine mark at its cognate motif throughout the host genome. This epigenetic signature distinguishes self from non-self. Phage DNA entering the cell is not methylated at these motifs and is therefore flagged as foreign, triggering exclusion/restriction. Because NF042963 itself has no nuclease, the model predicts (as in BREX) that the actual DNA-targeting/killing step is carried out by a separate component of the defense system, while NF042963 provides the essential **methylation-based recognition** layer. The **DUF1156 domain and C-terminal extension** most plausibly provide **DNA target recognition and/or complex assembly** — the accessory functions required to methylate a specific motif and to engage partner effectors.

Where this places the family among named defense strategies

Strategy	Core enzyme	Nuclease in same protein?	Match to NF042963
Type I/II/III R-M	MTase + endonuclease	Yes	Partial — has MTase, but **no nuclease** ✗
BREX	BrxX/PglX **SAM MTase**	No (effector in *trans*)	**Strong** ✓
DISARM	DrmMII **methylase**	No (multi-gene)	**Strong** ✓
Abortive infection	diverse toxins/ effectors	—	Weak (no toxin/effector fold) ✗
Nucleic-acid degradation	nuclease	Yes	**Excluded** — no nuclease fold ✗
Nucleotide signalling	cyclase / TIR	—	Excluded (no such fold) ✗

The architecture — a large, multidomain, SAM-dependent DNA methyltransferase with no accompanying nuclease — aligns most closely with **BREX/DISARM-type modification-dependent exclusion**.

Evidence Base

Structure- and sequence-derived evidence (this work)

- **Foldseek (AFDB-SwissProt + PDB100):** top hits at probability 1.0 are all m4C DNA methyltransferases — M.MjaII (Q58843), M.MvaI (P14244), and cyanobacterial M.AvaI (P0A461/P0A462), each EC 2.1.1.113.
- **InterPro:** IPR029063 (SAM-dependent MTase superfamily) at residues 157–287; DUF1156 = IPR009537/PF06634 at residues 30–90 (separate accessory domain).
- **AlphaFold model AF-A0A3B7MFS0-F1** (mean pLDDT 86.8): motif I (F233–G237) and catalytic DPPY (D617–Y620) converge in one pocket (G235–D617 Cα = 8.1 Å).

Supporting literature

PMID: 39979294 — *Molecular basis of foreign DNA recognition by BREX anti-phage immunity system.* This paper establishes the core principle underlying the proposed mechanism: "**Anti-phage systems of the BREX (BacteRiophage EXclusion) superfamily rely on site-specific epigenetic DNA methylation to discriminate between the host and invading DNA.**" It further specifies that "**defense and methylation require BREX site DNA binding by the BrxX (PglX) methyltransferase employing S-adenosyl methionine as a cofactor**." These statements provide direct precedent that a large, SAM-dependent DNA methyltransferase can be the mechanistic core of a validated anti-phage system that works by self/non-self methylation — exactly the architecture and cofactor dependence found for NF042963.

PMID: 29085076 — *DISARM is a widespread bacterial defence system with broad anti-phage activities.* This work shows that "**the B. paralicheniformis DISARM methylase modifies host CCWGG motifs as a marker of self DNA akin to restriction-modification systems.**" This is the exact modification-based, motif-specific self-marking mechanism proposed for the DUF1156 family, in the context of a multi-gene defense system where the methylase provides the recognition layer.

PMID: 32338761 — *Phage T7 DNA mimic protein Ocr is a potent inhibitor of BREX defence.* This paper reinforces that BREX "self versus non-self differentiation requires methylation of specific asymmetric sites in host DNA by BrxX (PglX) methyltransferase," and that the phage counter-defense protein Ocr "physically associates with BrxX methyltransferase" — evidence that the methyltransferase is the functional heart of exclusion-type defense and a target for phage antagonists. It supports the view that NF042963's methyltransferase is the defensive engine.

PMID: 40921744 — *Multigenerational proteolytic inactivation of restriction upon subtle genomic hypomethylation in Pseudomonas aeruginosa.* This study underscores the general logic that methylation-based systems must methylate self DNA to avoid autoimmunity ("This post-translational regulation prevents self-DNA targeting, which is a risk due to stable genomic hypomethylation"). It contextualizes why a dedicated, high-fidelity DNA methyltransferase (as found in NF042963) is central to modification-based defense.

Original phenotype reference

The family's anti-phage phenotype was established by **Gao et al. 2020, *Science* 369:1077 (PMID: 32855333)**, "*Diverse enzymatic activities mediate antiviral immunity in prokaryotes*," which validated protection against phage but did not resolve the molecular mechanism. This report supplies the mechanistic assignment that the screen left open.

Limitations and Knowledge Gaps

1. **No wet-lab confirmation of methyltransferase activity.** The assignment is entirely computational (sequence motifs + AlphaFold model + Foldseek). Enzymatic activity, cofactor binding, and the modified base (m4C vs. an unexpected m6A) have not been measured biochemically.
2. **Recognition sequence unknown.** The specific DNA motif methylated by the family is not determined. Without it, the "self-mark" cannot be mapped, and the trans-acting effector (if any) cannot be predicted from motif context.
3. **The trans-effector is inferred, not identified.** The BREX/DISARM analogy predicts a separate nuclease/effector encoded elsewhere in the operon. The genomic neighborhood of NF042963 members was not analyzed here; whether a cognate effector exists (BREX-like) or whether methylation alone suffices (a pure exclusion/replication-block mechanism) is unresolved.
4. **Function of the DUF1156 and C-terminal modules is inferred.** Foldseek found no structural homolog for either accessory region, so their assignment as target-recognition/scaffolding is by analogy to MTase architecture rather than by direct evidence. They could instead mediate partner-protein contacts or signal transduction.
5. **m4C vs. m6A specificity carries residual uncertainty.** Although the top Foldseek hits are all m4C enzymes and no m6A enzyme outranked them, both use the DPPY/amino-MTase chemistry; discriminating N4-cytosine from N6-adenine definitively requires biochemical mapping of the modified base.
6. **Single representative analyzed.** Only A0A3B7MFS0 was examined structurally. Conservation of the DPPY (motif IV) and FxGxG (motif I) motifs and domain architecture across all ~151 family members was not verified by a family-wide multiple sequence alignment.

Proposed Follow-up Experiments / Actions

1. **Decisive confirmatory experiment (catalytic-dead mutant).** Construct **D617A** and/or **Y620A** (DPPY motif IV) point mutants and a motif-I control (e.g., G235A). Show in the native or a heterologous host that these mutants **abolish both (a) DNA N4-cytosine methylation and (b) phage protection**, while the wild type retains both. Loss of protection upon catalytic inactivation would establish that methyltransferase activity is required for defense — the single cleanest test of the model.
2. **Direct biochemistry.** Express and purify the recombinant protein (or its central module); assay SAM-dependent DNA methyltransferase activity in vitro (^{3}H-SAM transfer into a defined dsDNA substrate, or enzyme-coupled assays) and confirm the **modified base is N4-methylcytosine** by LC-MS/MS of digested DNA.
3. **Recognition-motif mapping.** Use **SMRT (PacBio) or Nanopore methylation sequencing** of host DNA from cells expressing the wild-type vs. catalytic-dead protein to identify the cognate methylation motif and confirm genome-wide self-marking.
4. **Operon / trans-effector analysis.** Examine the genomic neighborhoods of NF042963 members for co-encoded nuclease/effector or ATPase genes (BREX/DISARM signature genes), to determine whether defense is exclusion-only or requires a trans-acting effector.
5. **Family-wide conservation check.** Build a multiple sequence alignment of all ~151 NF042963 members and confirm strict conservation of motif I (FxGxG) and catalytic motif IV (DPPY), plus the accessory DUF1156 domain, to verify the mechanism generalizes beyond the single representative.
6. **Phage counter-defense test.** Assess whether the T7 DNA-mimic protein **Ocr** (a known BREX/R-M inhibitor) neutralizes NF042963-mediated defense, which would further tie the family to the BREX/exclusion class.

Conclusion

The NF042963 / DUF1156 anti-phage family is, mechanistically, a **site-specific SAM-dependent DNA cytosine-N4 methyltransferase (m4C; EC 2.1.1.113)**. Structure-based fold assignment identifies a single intact catalytic module (β-class amino-MTase) with a fully preserved SAM-binding motif I (FxGxG, F233–G237) and catalytic motif IV (DPPY, D617–Y620) forming one Rossmann-fold active site; Foldseek's top matches at probability 1.0 are all N4-cytosine DNA methyltransferases. The protein contains no nuclease and no other catalytic fold, and the DUF1156 domain plus C-terminal extension are accessory (recognition/scaffolding). The family therefore most plausibly confers phage defense by **methylation-based self versus non-self discrimination in the BREX/DISARM/restriction–modification mould** — writing an N4-cytosine self-mark on host DNA so that unmethylated invading phage DNA is excluded/restricted, with any effector step supplied in trans. **Confidence is high for the enzymatic identity and moderate-to-high for the modification-dependent exclusion mechanism.** The definitive confirmation is a DPPY catalytic-dead mutant (D617A/Y620A) that abolishes both methylation and phage protection.

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